Lab 09

Migrating APIs using the API Controller

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs and paths in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **`<GW_HTTPS_PORT>` / `<GW_HTTP_PORT>`** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **`<PROD_HTTPS_PORT>`** – HTTPS port of the second (production) API Manager: 9445 with port offset 2.
- **`<APIM_HOME>`** – the folder where API Manager is extracted.
- **`<APICTL_DIR>`** – the folder that contains the apictl program after you extract it, for example ~/apictl (Windows: C:\apictl).

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue. The -k option in the apictl commands skips the certificate check for the same reason.

The commands use `apictl` and assume its folder is on your PATH (see the Creating an API Project lab). If it is not, run the commands from inside the apictl folder as `./apictl` (Windows: `apictl.exe`).

Training Objective

Learn how to use the API Controller to move an API between two API Manager environments of the same version: export the PizzaShack API (created in an earlier lab) from dev as a zip file, and import it into production.

Business Scenario

PizzaShack maintains its APIs in several environments. APIs are built and tested in the development environment and then moved, with the same definition, to the production environment.

High-Level Steps

- Prerequisites: a second (production) API Manager
- Set up the API Controller environments
- Export the PizzaShack API as a zip file
- Import the API into production

Prerequisites

You need two API Manager 4.7.0 instances: the one used in the previous labs as the dev environment (no port offset), and a second one as the production environment with port offset 2. To set up the second one:

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1. Extract a second copy of the API Manager 4.7.0 zip into a different folder.

2. In its `repository/conf/deployment.toml`, add this line under the `[server]` section:

```
offset = 2
```

3. Start it from its `bin` folder. Its portals are on port 9445 and its gateway on 8245 (HTTPS). Running two instances needs about 4 GB more free RAM.

   macOS / Linux

```
sh api-manager.sh
```

   Windows

```
api-manager.bat
```

   If a port is already in use on your machine, use another offset and adjust the URLs (HTTPS port = 9443 + offset).

You also need the PizzaShack API created and deployed in the dev environment in the Creating and Publishing an API lab.

Part 1 – Set Up the API Controller

4. apictl must be installed and on your PATH. If you have not done this yet, follow **Part 1 – Install and Initialise apictl** in the Creating an API Project lab. Check it with:

```
apictl version
```

5. List the available commands (optional):

```
apictl --help
```

6. Add the dev environment. Skip this if you added it in the Creating an API Project lab: adding an environment that already exists fails.

```
apictl add env dev --apim https://<APIM_HOST>:<HTTPS_PORT>
```

   Example:

```
apictl add env dev --apim https://localhost:9443
```

7. Add the production environment:

```
apictl add env production --apim https://<APIM_HOST>:<PROD_HTTPS_PORT>
```

   Example:

```
apictl add env production --apim https://localhost:9445
```

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8. List the environments and check that dev and production point to the right ports:

```
apictl get envs
```

Part 2 – Export the API from Dev

9. Log in to the dev environment (apictl asks for the password):

```
apictl login dev -u admin -k
```

10. List the APIs in the dev environment. Note the exact **NAME**, **VERSION** and **PROVIDER** of the PizzaShack API:

```
apictl get apis -e dev -k
```

11. Export the API as a zip file. `-r` is the provider: the user who created the API (`creator` in the earlier lab), not the role name. `--rev` is the revision number shown on the API's **Deployments** page; use `--latest` instead to export the latest revision.

    Command format – macOS / Linux

```
apictl export api -n <API-name> -v <version> \
  --rev <revision-number> -r <provider> -e dev -k
```

    Command format – Windows

```
apictl export api -n <API-name> -v <version> ^
  --rev <revision-number> -r <provider> -e dev -k
```

    Sample command:

```
apictl export api -n PizzaShack -v 1.0.0 --rev 1 -r creator -e dev -k
```

    The zip file is saved in the apictl home folder (the command prints the exact path), for example `~/.wso2apictl/exported/apis/dev/PizzaShack_1.0.0.zip` (Windows: `%USERPROFILE%\.wso2apictl\exported\apis\dev`).

Part 3 – Import the API into Production

12. Log in to the production environment:

```
apictl login production -u admin -k
```

13. Import the API. A path that starts with `dev/` is looked up in the apictl export folder; you can also give the full path printed by the export command.

    Command format – macOS / Linux

```
apictl import api -f <path-to-exported-zip> -e production -k \
  --rotate-revision
```

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Command format – Windows

```
apictl import api -f <path-to-exported-zip> -e production -k ^
  --rotate-revision
```

Sample command:

```
apictl import api -f dev/PizzaShack_1.0.0.zip -e production -k --rotate-revision
```

Expected output: `Successfully imported API`. Check it with:

```
apictl get apis -e production -k
```

Notes

- To import the same API again (for example after re-exporting), add `--update`. Without it the import fails with `409 The API already exists`.
- The API keeps its provider (`creator`) even though that user does not exist in production. To make the importing user the provider, add `--preserve-provider=false`.
- The API also keeps the dev back-end endpoint. In real migrations, use an environment parameters file (the apictl params file) to set production endpoints, certificates and other values during import.

Expected Outcome

The PizzaShack API is now in the production environment (PUBLISHED and deployed). It is visible in the production Developer Portal at `https://<APIM_HOST>:<PROD_HTTPS_PORT>/devportal` and can be invoked through the production gateway.

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